5 The lines l_1 and l_2 have equations $\mathbf{r} = \mathbf{i} + 4\mathbf{j} - \mathbf{k} + \lambda(\mathbf{j} - 2\mathbf{k})$ and $\mathbf{r} = -3\mathbf{i} + 4\mathbf{j} + \mu(\mathbf{i} + 2\mathbf{j} + \mathbf{k})$ respectively.

(a) Find the shortest distance between l_1 and l_2. [5]

The plane Π_1 contains l_1 and is parallel to l_2.

(b) Obtain an equation of Π_1 in the form $px + qy + rz = s$. [2]

(c) The point $(1,1,1)$ lies on the plane Π_2.

It is given that the line of intersection of the planes Π_1 and Π_2 passes through the point $(0,0,2)$ and is parallel to the vector $\mathbf{i}+4\mathbf{j}-3\mathbf{k}$.

Obtain an equation of Π_2 in the form $ax+by+cz=d$. [3]